

* Séries alternadas:

* Lévêrs.

$$\begin{cases} \rho_{n+1} \leq \rho_n \\ \lim_{n \rightarrow \infty} \rho_n = 0 \end{cases}$$

$$(1) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = (-1)^1 \cdot \frac{1}{1} = \frac{1}{2} = \frac{1}{2} \quad \frac{1}{2} = \frac{1}{2} \quad \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad * \text{Converge}$$

$$(2) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} = \frac{1}{2 \cdot 1 - 1} = \frac{1}{1} = 1 \geq \frac{1}{2 \cdot 3 - 1} = \frac{1}{5} \quad \lim_{n \rightarrow \infty} \frac{1}{2n-1} = 0 \quad * \text{Converge}$$

$$(5) \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2 + 1} \rightarrow \frac{\rho_2}{5} \leq \frac{\rho_3}{10} \quad * \text{Não atende o Lévêrs} \quad \therefore \text{Diverge}$$

$$(7) \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \rightarrow \frac{\rho_2}{\sqrt{2}} > \frac{\rho_3}{\sqrt{3}} \quad \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0 \quad * \text{Converge}$$

$$(9) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(n+1)!} \rightarrow \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+2)!} = \lim_{n \rightarrow \infty} \frac{1}{n+2} = 0 \quad * \text{Converge}$$

$$(11) \sum_{n=1}^{\infty} \frac{(-1)^n}{e^n} \rightarrow \frac{1}{e} \leq \frac{1}{e^2} \quad \lim_{n \rightarrow \infty} \frac{1}{e^n} = 0 \quad * \text{Converge}$$

$$(13) \sum_{n=1}^{\infty} \frac{\sin(2n-1)\pi}{2} \rightarrow \{1, -1, 1, -1, \dots\} \rightarrow \sum_{n=1}^{\infty} (-1)^{n+1} \quad \lim_{n \rightarrow \infty} 1 = 1 \quad \therefore \text{Diverge}$$

$$(15) \sum_{n=1}^{\infty} \frac{1}{n} \sin(2n-1)\pi = \sum_{n=1}^{\infty} \frac{1}{n} \cdot (-1)^{n+1} \rightarrow \frac{1}{1} \geq \frac{1}{2} \quad \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \therefore \text{Converge}$$

$$(17) \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \rightarrow \left\{ \frac{1}{1}, \frac{1}{2}, \frac{1}{6}, \dots \right\} \quad \lim_{n \rightarrow \infty} \frac{1}{n!} = 0 \quad \therefore \text{Converge}$$

$$(19) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+2} \rightarrow \left\{ \frac{1}{3}, \frac{\sqrt{2}}{4}, \dots \right\}$$

$\rho_2 > \rho_1 \quad * \text{Diverge}$

$$(21) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{e^n - e^{-n}} \quad \lim_{n \rightarrow \infty} \frac{2}{e \cdot (1 - \frac{1}{e})} = 0 \quad \therefore \text{Converge}$$

* Se $\sum |a_n|$ converge, então $\sum a_n$ também converge.

L> O contrário não é válido!

* Convergência absoluta e condicional

- absoluta: Se $\sum |a_n|$ converge

- condicional: Se $\sum a_n$ converge, mas $\sum |a_n|$ diverge

→ Teste normal:

$$\begin{cases} \rho_{n+1} \leq \rho_n \\ \lim_{n \rightarrow \infty} \rho_n = 0 \end{cases}$$

Testar módulo primeiro, já saberemos depois se é possível decidir.

$$(23) \sum_{n=1}^{\infty} \frac{1}{(n+1)!} \quad (\text{use teste de módulo}) \rightarrow \text{Converge}$$

L> Converge absolutamente.

$$(25) \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \text{seu } p, p > 1 \rightarrow \text{Converge}$$

L> Testar o original

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \quad \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0 \quad \text{OK}$$

$\therefore$ Converge condicionalmente

* Verificar convergência

$$\frac{1}{\sqrt{n+1}} \leq \frac{1}{\sqrt{n}} \quad \text{OK}$$

$$(27) \sum_{n=1}^{\infty} \frac{1}{\ln(n)} \rightarrow \lim_{n \rightarrow \infty} \frac{1}{\ln(n)} = 0 \quad * \text{Inconclusivo}$$

* Comparação de limites: Séries para comparação: $\sum_{n=1}^{\infty} \frac{1}{n}$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{\ln(n)}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{\ln(n)} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n}} = \infty \quad * \text{Diverge.}$$

L> Como o do módulo diverge, o restante testar o série original e verificar se converge.

$$* \text{Teste de original: } \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\ln(n)}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\ln(n)} = 0 \quad \text{OK}$$

$$\frac{1}{\ln(n+1)} \leq \frac{1}{\ln(n)} \quad \text{OK}$$

$$(29) \sum_{n=1}^{\infty} \frac{1}{n^2 - 1} \rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^2 - 1} \rightarrow \lim_{n \rightarrow \infty} \frac{1}{6n^2} = \frac{1}{6} \quad * \text{Converge}$$

L> Converge absolutamente.

$$(33) \sum_{n=0}^{\infty} \cos n\pi \rightarrow \lim_{n \rightarrow \infty} |\cos n\pi| = \lim_{n \rightarrow \infty} 1 \quad * \text{Diverge}$$

$$\lim_{n \rightarrow \infty} \cos n\pi = 0 \quad \text{OK} \quad \rho_{n+1} \leq \rho_n \rightarrow \frac{|\cos(n+1)\pi|}{n+2} \leq \frac{|\cos n\pi|}{n+1}$$

$\therefore$ A série converge condicionalmente

Teorema de Cauchy: $|S - S_n| = |R_n| \leq \rho_{n+1}$

$$(39) \rho_{n+1} < \frac{1}{1000} \rightarrow \frac{1}{2 \cdot (n+1)^2 - 1} < \frac{1}{1000} \rightarrow 2(n+1)^2 - 1 > 1000 \rightarrow (n+1)^2 > \sqrt{500.5} \rightarrow n > 7,98 - 1 \rightarrow n > 6,98, \text{ ou seja, somar até o } 7^{\circ} \text{ termo.}$$

$$(41) \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \rightarrow \rho_{n+1} < \frac{1}{1000} \rightarrow \frac{1}{(n+1)!} < \frac{1}{1000} \rightarrow (n+1)! > 1000 \rightarrow \frac{7!}{(6+1)!} > 1000, \text{ ou seja, somar até o } 6^{\circ} \text{ termo.}$$

$$(43) \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+2)!} \rightarrow \rho_{n+2} < \frac{1}{1000} \rightarrow (2(n+2)+2)! > 1000 \rightarrow (2 \cdot (2+2)+2)! > 1000 \text{ somar até o } 2^{\circ} \text{ termo}$$

$$(45) \sum_{n=0}^{\infty} \frac{(-1)^{n+2}}{n \cdot 2^n} \rightarrow \rho_{n+1} < \frac{1}{1000} \rightarrow \frac{1}{(n+1) \cdot 2^{n+1}} < \frac{1}{1000} \rightarrow (n+1) \cdot 2^{(n+1)} > 1000 \rightarrow (1+1) \cdot 2^1 > 1000 \quad \therefore \text{somar até o } 1^{\circ} \text{ termo.}$$